5. RESCUESAT - Decentralized Disaster Damage Assessment

The RESCUESAT initiative aims to overcome the slow tasking times and high costs of traditional satellite imagery by deploying a rapid-response Earth observation network. It provides high-resolution imagery within four hours of a natural disaster, using blockchain technology to generate verifiable damage assessments that facilitate instant insurance claims and more efficient aid distribution.

Mission Brief

Design a rapid-response Earth observation satellite that provides high-resolution imagery within 4 hours of earthquakes/floods/hurricanes and uses blockchain to create verifiable damage assessments for instant insurance claims and aid distribution. Traditional satellites take days to task and imagery is expensive/controlled.

Satellite Requirements

- **Orbit:** LEO 500 km, high-revisit constellation (8-12 satellites)
- **Coverage:** Any point on Earth <4 hours after disaster alert
- **Payload:** RGB camera (5m resolution) + SAR radar (all-weather, 10m resolution)
- **Agility:** Rapid retargeting (±45° off-nadir pointing in 2 minutes)
- **Power:** 60W average (high-power SAR radar)
- **Mass:** <100 kg per satellite (custom smallsat design)
- **Comms:** X-band high-speed downlink (100 Mbps)

Web3 System

Data Flow: Disaster occurs (USGS earthquake alert / NOAA hurricane) → Smart contract automatically tasks nearest satellite → Satellite captures imagery within 4 hours → AI analyzes damage (building collapse, flooding extent) → Results hashed to blockchain → Insurance companies query damage at specific addresses → Instant claims approval for verified damage → Aid organizations access data for free

Key Smart Contracts:

1. **DisasterAlert** - Monitors external APIs (USGS, NOAA), auto-tasks satellites based on severity
2. **ImageryRegistry** - Stores IPFS hashes of pre/post-disaster imagery with geographic bounds
3. **DamageAssessment** - AI-generated damage scores per building/area, cryptographically signed
4. **InsuranceClaims** - Property owners submit claims, contract verifies damage at GPS coordinates
5. **AidDistribution** - NGOs access damage maps, smart contract tracks aid delivery to avoid duplication

Round 1 constraints:

1. Mission Objectives & Success Criteria

Every design must start with the **"Why."** In a hackathon, your mission objective isn't just a goal; it's a boundary.

- **Operational Lifetime:** How long must the satellite survive? (e.g., 6 months for a tech demo vs. 5 years for a weather satellite).
- **Target Orbit:** This dictates almost everything else.
    - **LEO (Low Earth Orbit):** Lower latency, higher resolution, but high atmospheric drag.
    - **GEO (Geostationary Orbit):** Constant coverage of one area, but extremely high launch costs and radiation.
- **Data Latency:** Does the data need to be "real-time" (for disaster relief) or can it be stored and forwarded once a day?

2. Payload Constraints

The payload is the "customer" of the satellite; everything else (power, thermal, comms) exists just to keep it happy.

- **SWaP-C:** This is the golden rule of satellite design: **S**ize, **W**eight, **a**nd **P**ower - **C**ost.
- **Mass & Volume:** Most hackathons use the "U" standard (1U = 10x10x10 cm). You'll likely be restricted to a 3U or 6U form factor.
- **Duty Cycle:** Can the payload run 24/7, or does it need to shut down to recharge batteries while the satellite is in the Earth's shadow?

3. Budget & Resource Estimation

In a real-world scenario (and a good hackathon), "unlimited money" isn't a thing.

- **Total Cost of Ownership (TCO):** You must account for:
    1. **Development:** Components and labor.
    2. **Launch:** Often calculated per kilogram (e.g., $50,000/kg for LEO).
    3. **Operations:** Ground station fees to talk to your satellite.
- **COTS vs. Custom:** Using **Commercial Off-The-Shelf (COTS)** parts is cheaper and faster but might not be "space-hardened" against radiation compared to custom, radiation-hardened components.

4. Technical Constraints & Requirements

1. Rapid-Response Sensing (The "Quick Eye")

- **The Dual-Sensor Payload:** Unlike standard satellites that use one camera type, RESCUESAT combines **RGB (optical)** for clear visual photos with **SAR (Radar)**. This is essential because disasters like hurricanes or floods often involve heavy cloud cover that blinds normal cameras; SAR "sees" through clouds and smoke to map damage 24/7.

- **Agile Tasking:** Traditional satellites are slow to turn. Our satellites feature **±45° off-nadir pointing**, meaning they can "tilt" toward a disaster zone in under 2 minutes as they pass by, rather than waiting to be directly overhead.
- **4-Hour Revisit:** A 12-satellite constellation ensures that no matter where a disaster strikes, a "Rescuer" satellite is in position to capture imagery within 240 minutes.

2. AI-Driven Damage Assessment

- **On-Chain Truth:** Instead of humans manually counting destroyed buildings, **Edge-AI** on the satellite compares "Pre-Disaster" photos with "Post-Disaster" imagery.
- **Verifiable Scores:** The AI generates a damage score for every GPS coordinate. This data is **hashed** (securely locked) onto the blockchain, ensuring that neither insurance companies nor governments can alter the findings to avoid payouts or hide the scale of a crisis.

3. Instant Insurance & Aid (The "Paymaster")

- **Parametric Claims:** We use **Smart Contracts** to eliminate the claims process. If the satellite verifies that "Building A" at "GPS X" has collapsed, the **InsuranceClaims** contract triggers an instant payout to the owner's wallet.
- **Coordinated Relief:** The **AidDistribution** contract creates a live "heatmap" of the most affected areas. NGOs can see where aid is flowing in real-time, preventing "aid duplication" (where one area gets too many supplies while another gets none).
- **Automated Tasking:** The system is "hands-off." It monitors global alert feeds (like USGS for earthquakes). When a major event occurs, the **DisasterAlert** contract automatically "tasks" (orders) the nearest satellite to zoom in, bypassing manual bureaucracy.

4. Technical & Mission Feasibility

- **Custom Smallsat Design:** By keeping the mass **<100 kg**, we can launch these in clusters on a single rocket, drastically lowering the cost per satellite.
- **High-Speed Link:** Disaster data is urgent. We use **X-band comms (100 Mbps)** to beam high-resolution damage maps to ground stations in minutes, not hours.
- **Decentralized Access:** While insurance companies pay for the data, aid organizations get access for **free**, ensuring that the mission serves humanity before profit.